

Profile

NTUST Chemical Engineering M.S. candidate holding three environmental specialist licences (wastewater, waste, indoor air), OSH exams scheduled; hands-on with hazardous materials (lithium metal, organic-solvent electrolytes, faulty Li-SOCl₂ cells), lab safety and thermal hazard analysis (TGA / DSC).

3	NT\$ 2M+	6	3 +2
Battery-analysis techniques run in the industry project: EIS, OCV drip, X-ray CT	Precision equipment procured & accepted	Labs across 3 departments served with TGA testing	3 environmental licences held, 2 OSH exams scheduled

Experience

2025 – Present	Sustainable Electrochemical Energy Development (SEED) Center, NTUST Lab Equipment Procurement & Asset Maintenance Manager Procurement (>NTD 2M): Led procurement of glovebox O₂ sensors, a DSC press set with annual maintenance contract, a hydraulic electrode punch, vacuum pumps, 10 Eppendorf pipettes and low-temperature baths. Lab safety: monitor glovebox O₂ / moisture levels and handle sensor faults; manage and inspect gas cylinders, run lab safety walk-throughs and brief newcomers on safety; helped with on-site response to incidents such as a cut from broken glassware and a glovebox catalyst-canister leak. Hazardous materials: handle lithium metal and assemble coin cells in a glovebox, prepare ether / carbonate organic-solvent electrolytes, and dismantle and handle faulty Li-SOCl₂ cells. TGA testing service: Ran TGA measurements and data interpretation for graduate students from 6 labs across Chemical, Mechanical and Materials Engineering at NTUST. Instrument maintenance: Responsible for maintenance and calibration of the key thermal analysis instruments, TGA Q500 and DSC 2500.
Summer 2022	New Taipei City Dept. of Urban Development Summer Intern, Housing Development Section Reviewed rental subsidy applications and indexed confidential records in government databases; handled frontline public inquiries, building strong familiarity with administrative workflows and SOPs.
2021 – 2024	Tunghai University Student Housing / GEPT Resident Assistant (2 yrs) / Exam Staff (part-time) Oversaw safety and daily order in a 100-resident dormitory, mediating resident disputes and liaising with university administration; as part-time exam staff, handled test-paper check-in with zero errors.

Research & Industry

Long-Life Li Battery Evaluation for Smart Water Meters

Partner: a smart-water-meter battery maker **Project lead**

Screening methodology Designed full-stage test protocols, experimental workflows and statistical validation models.	Advanced analysis EIS, OCV drip, self-discharge / lifetime estimation, accelerated aging and X-ray CT internal structure analysis.
Database & SOP Established the test database and standardized high-throughput screening procedures (SOP).	Training & tech transfer Trained QA and R&D teams to independently run large-batch screening and data interpretation.

Nature Energy Journal submission — battery thermal stability study Status **Under review**

Co-author Responsible for TGA thermal measurement

Performed TGA sample preparation, measurement setup and data interpretation for battery material samples, providing thermal stability data supporting the manuscript's conclusions.

Conferences & Training

Presenter	2026 TwICHe Annual Meeting Title: <i>Weakly Solvating Allyl Ether Electrolytes for Subzero Li SPAN Batteries: Solvation Structure, Transport Properties, and Separator Wettability</i>
Vendor seminar	TA Instruments ARES-G3 launch & applications forum 2026 / 07 / 17
Vendor lecture	TA Instruments TGA lecture, Taiwan 2026 / 06 / 03

EHS Licenses

✓	Dedicated Wastewater and Sewage Treatment Specialist (Class B) Certified 2026
✓	Indoor Air Quality Maintenance Management Specialist Certified 2026
✓	Waste Disposal Technician (Class A) Certified 2026
○	Occupational Safety and Health Management Personnel (Level B) 2026 / 11 · mid-Nov exam scheduled
○	Occupational Safety Management Specialist (Level A) 2027 / 03 · exam scheduled

Education

2025 – 2027 (expected)	National Taiwan University of Science and Technology M.S. in Chemical Engineering Focus on low-temperature electrolytes and battery aging. Coursework: Nanomaterials and Applications (lithography, nanoimprint & self-assembly), Process Safety & Radiation Protection.
2020 – 2024	Tunghai University B.S. in Chemical & Materials Engineering Focus on unit operations & process control (Aspen Plus V14 / Aspen Dynamics simulation).

Skills & Instruments

Electrochemistry & batteries Impedance spectroscopy (EIS) Battery aging analysis Reliability analysis Low-temp electrolytes
Characterization Thermal analysis (TGA / DSC) Synchrotron (TLS20A / TPS27A / 45A) Rheology (ARES-G3) Viscometer & densimeter (DMA5002)
Process & operations Lithography (nanoimprint / self-assembly) Equipment procurement & maintenance
Software Aspen Plus V14 Aspen Dynamics Microsoft Office OriginPro TA Universal Analysis TA TRIOS
Instruments TGA Q500 Maintenance DSC 2500 Maintenance ARES-G3 Operation Glovebox O ₂ sensor Procurement DSC sample press Procurement + service Hydraulic electrode punch Procurement Vacuum pump Procurement Eppendorf Pipette Procurement Low-temperature bath Procurement